



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

INDIVIDUAL TASK SESSION 20252026 SEM 01	
STUDENT'S NAME	GOE JIE YING
MATRIC NO.	A23CS0224
IC NO.	040622-01-0526
EMAIL	goeying@graduate.utm.my
YEAR/PROGRAME	3/SECPH
FACULTY	FACULTY OF COMPUTING
SUBJECT CODE	GLRT0060
SUBJECT TITLE	SAFETY AND HEALTH OFFICER INTRODUCTORY COURSE
SECTION	1
LECTURER'S NAME	ASSOC. PROF. SR. DR MOHD SAIDIN BIN MISNAN
DATE OF SUBMISSION	30 NOVEMBER 2025

Task 1 Safety (Computer Engineering)

Safety means managing electrical and static hazards, following ergonomic practices, and ensuring cybersecurity and system safety protocols are properly implemented. It also involves staying aware of potential hardware and software risks to prevent accidents, data loss, or system failures.

Task 2 Accident Prosecution Cases

3 accident prosecution cases related to computer engineering:

1. QTS Data Center, Fayetteville, Georgia, USA (June 2025)

A worker at the QTS Data Center in Fayetteville, Georgia was fatally electrocuted on June 28, 2025. He was repairing a light fixture while elevated on a scissor lift and came into contact with live wiring, reportedly due to improper grounding or a faulty temporary power setup. He died several days after the incident.

2. Telecommunications Technician, Udon Thani, Thailand (August 2025)

A telecommunications and internet service technician in Udon Thani, Thailand died in August 2025 after falling from a high voltage pole while repairing internet cables. Reports indicate he climbed the pole using a ladder and suffered a fatal fall. Investigations noted that the cause was a fall from height rather than electrocution.

3. Eagle's Nest Skywalk Langkawi, Malaysia (October 2024)

A maintenance technician at the Eagle's Nest Skywalk in Langkawi, Malaysia died in October 2024 after falling from a forty metre high deck while carrying out technical maintenance. The incident triggered workplace safety investigations by local authorities.

Task 3 Safety and Health Officer (SHO)

There are 9 duties of SHO listed in the regulation.

1. Advisor
2. Inspector or
3. auditor
4. Coordinator
5. Investigator
6. Organizer
7. Secretary
8. Reporter
9. Magistrate
10. Quality checker

Task 4 OSHA 1994

Duties of employer (Section 15, OSHA 1994):

1. To provide and maintain a safe plant and working system;
2. To put in place procedures so as to ensure safety and health in the usage, handling, storage and transportation of the plant and supplies;
3. To provide information, instruction, training and supervision for new and old employees;
4. To provide and maintain a working environment that is safe and free from health hazards; and
5. To provide the necessary amenities for the welfare of those who are working

Court Case:

In 2021, the Sessions Court fined a Japanese manufacturing company RM30,000 for negligence, which resulted in its employee's death. The victim was killed while repairing the company's machine, two years earlier. Based on the charge sheet, the company was accused of failing to provide training to its employees in the injection molding process, which resulted in the employee's death. The employer was charged under Section 15(2)(c) of the Occupational Safety and Health Act 1994, which carries a maximum of RM50,000 fine and two years of imprisonment.

Task 5 OSHA 1994

Duties of employee (Section 24, OSHA 1994):

1. To take reasonable care for the safety and health of himself and of other persons who may be affected by his acts or omissions at work;
2. To co-operate with his employer or any other person in the discharge of any duty or requirement imposed on the employer or that other person by this Act or any regulation made thereunder;
3. To wear or use at all times any protective equipment or clothing provided by the employer for the purpose of preventing risks to his safety and health; and
4. To comply with any instruction or measure on occupational safety and health instituted by his employer or any other person by or under this Act or any regulation made thereunder.
5. Be reasonably attentive so as to avoid injury onto themselves as well as others while carrying out their work activities

Court Case:

On 8 January 2025 at the Ipoh Magistrate's Court, an employee was charged under Section 24(1)(a) of the Occupational Safety and Health Act 1994 for failing to take reasonable care for his own safety while performing his duties. He was found to have acted in a manner that exposed himself and others to potential harm in the workplace. The court imposed a RM800 fine for him.

Task 6 OSHA 1994

5 regulations under OSHA 1994 :

1. Occupational Safety and Health (Employer's Safety and Health General Policy Statements)(Exception) Regulations 1995;
2. Occupational Safety and Health (Control of Industrial Major Accident Hazard) Regulations 1996;
3. Occupational Safety and Health (Safety and Health Committee) Regulations 1996;
4. Occupational Safety and Health (Classification, Packaging and Labeling of Hazardous Chemicals) Regulations 1997;
5. Occupational Safety and Health (Safety and Health Officer) Regulations 1997

Task 7 OSH related Guidelines

5 OSH related guidelines related to Computer Engineer:

1. Maintain neutral postures with proper neck, spine, elbow, shoulder, and wrist alignment to prevent musculoskeletal and repetitive strain injuries while using the computer.
2. Always power off and unplug computers before servicing, and use an anti-static wrist strap to prevent shocks and electrostatic damage.
3. Handle tools carefully, pulling connectors by their handles rather than cables to avoid equipment damage and electrical hazards.
4. Keep the workspace clear of liquids, food, and clutter, and ensure cables and pins are connected properly without excessive force.
5. Identify workplace hazards, understand emergency procedures, and use PPE correctly to minimize risks during incidents.

Task 8 Accident Causation

3 Fatal Accident cases for engineering:

1. In Penang, a maintenance technician tragically died when his head became trapped between the doors of a lift while performing routine servicing. The lift reportedly moved unexpectedly during inspection, causing fatal injuries. The incident prompted an investigation by DOSH under the Occupational Safety and Health Act 1994 (OSHA 1994) to determine whether the employer had failed to implement proper safety procedures, including lockout-tagout protocols and machine guarding, which could have prevented the accident.
2. On Langkawi, a technician fell to his death from a 40-metre-high glass-floor skywalk while performing technical access work on the elevated structure. Reports indicate he slipped from a metal safety barrier, and there were questions about whether adequate fall-protection equipment and procedures were in place. DOSH launched an inquiry to examine compliance with OSHA 1994 requirements, focusing on working-at-height safety measures, supervision, and risk assessments.

3. In Bintulu, Sarawak, a fatal accident occurred at a polysilicon plant when a hazardous gas release during a chemical-process failure led to the deaths of technical staff on duty. The release involved hydrogen chloride, and investigations revealed gaps in process safety management, emergency response planning, and system monitoring. The case highlights the importance of strong organizational safety systems, hazard identification, and engineering controls to prevent fatalities in complex industrial environments.

Task 9 Accident Theory

3 Accident Theories:

1. Domino Theory (Heinrich's Theory)

The Domino Theory explains that accidents occur through a sequence of five events, similar to falling dominoes: (1) social environment/ancestry, (2) a person's fault or characteristics, (3) unsafe acts or conditions, (4) the accident itself, and (5) injury. Heinrich emphasized that removing any one domino, especially unsafe acts to break the chain, preventing the accident from happening.

2. Human Factors Theory

The Human Factors Theory attributes accidents to human error resulting from factors such as fatigue, stress, overload, poor training, or inappropriate responses. It focuses on three main contributors: overload, inappropriate activities, and inappropriate responses. Prevention under this theory involves improving training, reducing stressors, ensuring proper workload, and designing work to reduce human mistakes.

3. Swiss Cheese Model (Reason's Model)

The Swiss Cheese Model views accidents as the result of multiple layers of defense failing at the same time. The procedures, training, equipment safeguards, and management systems layers each contain "holes," and when the holes align, an accident occurs. The model distinguishes between errors by frontline workers and underlying organizational weaknesses, stressing that both must be addressed to prevent incidents.

Task 10 Unsafe Act vs Unsafe Condition

Unsafe act vs unsafe condition at workplace:

1. Bypassing or removing safety devices (Unsafe Act)
2. Not reporting near-misses or potential hazards to supervisors (Unsafe Act)
3. Working under the influence of alcohol or drugs (Unsafe Act)
4. Slippery or wet floors with no warning signs (Unsafe Condition)
5. Exposed or frayed electrical wiring (Unsafe Condition)

Task 11 Safety Policy

Please identify UTM safety policy (from established sources)

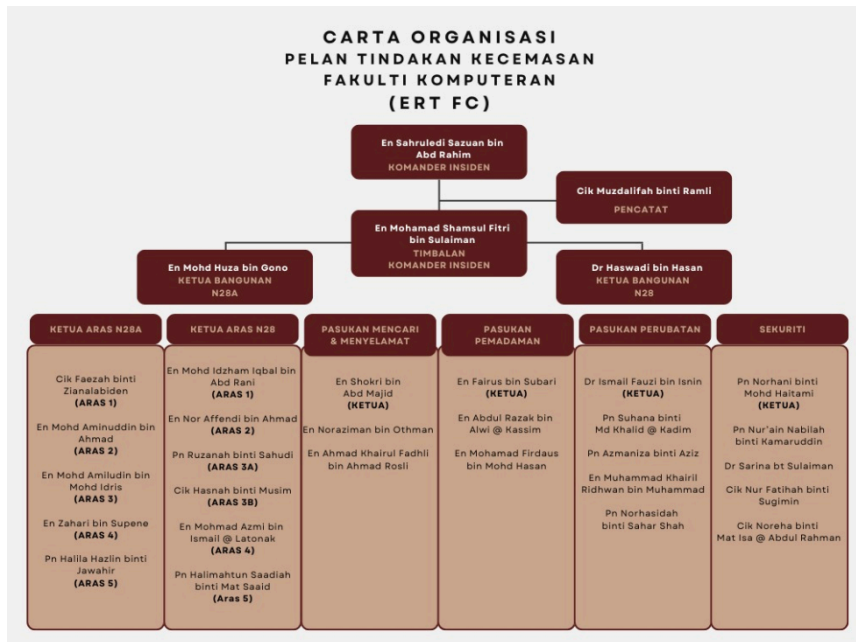
1. Occupational Safety, Heath and Environment signed by Prof. Dr. Mohd Shafry bin Mohd Rahim, Vice Chancellor Universiti Teknologi Malaysia on 9 January 2025.

Task 12 Safety Management Plan

1. Faculty/School Safety and Health Committee (JKKP/OSHEC)



2. Faculty/School Emergency Response Team (ERT)



3. UTM Emergency contact no.

**Occupational Safety, Health & Environment (OSHE) Office**
Pejabat Persekitaran, Keselamatan & Kesihatan Pekerjaan | UTM Johor Bahru

NOMBOR KECEMASAN

UMUM

POLIS

BOMBA

AMBULANS

JPAM

PUSAT RACUN NEGARA
(1-800-88-8099 / 012-4309499)



DALAMAN

BAHAGIAN KESELAMATAN 07-553 0014

PUSAT KESIHATAN UTM 07-553 0999

UNIT PERSEKITARAN, KESELAMATAN & KESIHATAN PEKERJAAN (OSHE) 07-553 1886

SEKITAR UTM

POLIS

Balai Polis Taman Universiti	07 - 520 3129
Balai Polis Skudai	07 - 556 1222
Balai Polis Kangar Pulai	07 - 527 1255
Balai Polis Senai	07 - 599 1222
Balai Polis Pekan Nenas	07 - 699 1222

BOMBA

Balai Bomba Pulai, Taman Universiti	07 - 520 4144
Balai Bomba Larkin	07 - 224 1222
Balai Bomba Pekan Nenas	07 - 699 4444
Balai Bomba Kulai	07 - 663 4444

HOSPITAL

Pusat Kesihatan Taman Universiti	07 -521 6800
Hospital Sultan Aminah	07 - 223 1666
Hospital Kulai	07 - 662 3333
Hospital Sultan Ismail	07 - 356 5000
Puteri Specialist Hospital	07 - 225 3222

JABATAN BERKAITAN

Jabatan Keselamatan & kesihatan Pekerjaan Johor	07 - 223 7071
Jabatan Alam Sekitar Johor	07 - 235 6041

<https://www.utm.my/oshe/>

Task 13 Spot the Hazard on slide 17



1. A worker is lifting items under poorly stacked boxes, creating a risk of the boxes falling and injuring someone.
2. Wires are scattered on the ground near the ladder; if someone steps on them and pulls the wires, the ladder could fall and injure the person on it.
3. A floor opening or drain cover is not properly closed, creating a risk of someone falling in.
4. A worker carrying a long wooden pole has no protective measures and may accidentally hit others.
5. A worker is standing on unstable, poorly stacked boxes and is not wearing any PPE.
6. Four heavy metal pipes are being lifted without proper securing, and there are workers standing directly underneath, creating a serious falling-object hazard.
7. The floor is wet and slippery, increasing the risk of slips and falls.